Lesson 4-4 · Period 1

Adding & Subtracting Rational Expressions + Resistance DOK-3 (Q#15 Prep)

Klimsara-adapted · Student-centered · No Blooket

Do Now — Critique & Explain

Algebra 2 · Lesson 4-4 · Single-Period Quick

DOK 1 **5 min** *bridge from 4-3*

Do Now (4-4-savvas-model-discuss-lesson-4-4-launch) · Common denominator conflict Teo claims that a common denominator of $\frac{1}{2}+\frac{1}{3}+\frac{1}{9}$ is $2+3+9=14$. Evander claims it is $2\cdot 3\cdot 9=54$.

- **A.** Is either student correct? Explain using the word *common denominator*.
- **B.** Find the sum. Explain your method.
- **C.** Timothy says: multiply denominators together — always quickest. Do you agree?

Launch — Rules + Example 3 + Example 4 (teacher models) Algebra 2 · Lesson 4-4 · Single-Period Quick

DOK 1–2 | 13 min | *teacher models*

Rules — Add/Subtract Rational Expressions (keep printed)

1. LIKE denominators: combine numerators, keep denominator, simplify.
2. UNLIKE denominators: (a) factor each denominator, (b) find the LCM, (c) rewrite each fraction with LCM as denominator, (d) combine numerators, (e) simplify.
3. LCM = every DISTINCT factor at its highest power across both denominators.
4. Always check: can the final expression simplify by cancelling a common factor?

COMPRESSED: Teacher models Ex 3 (add unlike) AND Ex 4 (subtract unlike). Watch the minus sign on Ex 4 — it distributes across EVERY term.

DOK 2–3 | 32 min | *student work*

6 items in order. Plan ∼10 min for Practice #32 (Resistance). Wed 45-min variant: skip #16 + #26.

Try It 3 · Add unlike denominators Find the sum of two rational expressions with unlike denominators. Factor first; build the LCM.

Try It 4 · Subtract unlike denominators Simplify. Watch: minus sign distributes across EVERY term of the second numerator.

Practice #12 · Like denominators **Generalize** — how is adding rational expressions similar to and different from adding rational numbers?

Share / Summary

DOK 2 | 5 min | *DOK-3 reveal + Q#15 bridge*

Sentence frame · Resistance (#32) "For the parallel circuit, $1/R_{\text{total}} = 1/R_1 + 1/R_2$. I found a common denominator of ______, added the reciprocals to get $1/R_{\text{total}} =$ _________, then flipped to get $R_{\text{total}} =$ ______. The ratio R_A/R_B simplified to ______ because ____________."

Bridge to Topic 4 LEHS Q#15 The resistance move — find LCM, combine reciprocals, form a single rational expression — is the structural reasoning tested on Topic 4 LEHS Q#15. You have now rehearsed it.

Exit Ticket — Summary Recap

Algebra 2 · Lesson 4-4 · Single-Period Quick

DOK 1–2 | 3 min | *not graded*

Complete on your exit ticket

1. To add rational expressions with unlike denominators, I must first find the ____________________ of the denominators.
2. Practice #32 (resistance) showed me that reciprocal-sum problems (like $1/R_1 + 1/R_2$) use the same algebraic steps as adding rational expressions because ____________________.
3. One biggest thing I learned today: ______________________________